

CPU pilot for spacetime anisotropy from Gaussian reference ancillas

Numerical methods and pilot report

August 26, 2026

Abstract

We implement the reference-qubit spacetime-correlation protocol of Zabalo *et al.* in the charge-conserving Gaussian adaptive circuit. A wall unit cell and a two-mode spectator reference are prepared in a maximally entangled Gaussian state, the canonical adaptive circuit continues, and the exact mutual information between two reference subsystems is evaluated from their reduced correlation matrix. The implementation passes analytic Bell-pair tests and a canonical-engine integration test. A CPU pilot at circumferences $L = 6, 8$ with four Born trajectories per configuration shows that the observable is feasible and has a long-lived signal. It does not yet calibrate the anisotropy: the distribution is rare-event dominated, the $L = 6$ crossing is only a candidate, and the $L = 8$ mean curve is nonmonotone because of one large trajectory. The pilot supports a next run with at least $S = 100$ trajectories per configuration and roughly $7L$ total cycles per anisotropy trajectory.

1 Conclusion first

Protocol correction. The completed run documented below used an explicit domain wall with `dw_truncation=False` and `meas_slab_only=False`. The target campaign is now locked to `dw_truncation=True` and `meas_slab_only=True`. The saved run remains a valid engineering test of Gaussian reference insertion, entropy evaluation, and statistical failure modes, but its candidate α is not a calibration of the support-terminated protocol. The target geometry requires a new convergence and anisotropy run; no result below is silently relabeled.

The correct reference observable can be implemented without leaving the Gaussian covariance/occupied-frame description. The pilot therefore resolves the earlier engineering question: reference ancillas are not blocked by the noninvertibility of individual measurement maps or by nonfinite many-body log singular values. They are ordinary spectator fermionic modes carried by the pure Gaussian frame.

The present numerical result is a *feasibility success* but not an accepted measurement of α . The strict analysis gives

$$\begin{aligned} L = 6 : \quad t_* &= 1.448, & \alpha_{\text{cand}} &= 1.163, \\ L = 8 : \quad t_* &\text{ unresolved.} \end{aligned}$$

For $L = 6$, only 75.6% of trajectory-bootstrap resamples contain a valid ordered crossing, and the earlier permissive 95% interval was broad. For $L = 8$, the temporal mean begins below the spatial target, rises at the second separation because of a single rare trajectory, and then falls. A late downward interpolation through that upward fluctuation is not a CFT matching point and is rejected.

2 CFT matching prescription

For a primary field of scaling dimension Δ on a cylinder of spatial circumference L , an anisotropic Euclidean coordinate is $z' = \alpha t + iy$. The plane-to-cylinder map gives the equal-time, half-circumference correlator and the equal-position temporal correlator as

$$g'_{\text{space}}(L/2, 0) = \left(\frac{\pi}{L}\right)^{2\Delta}, \tag{1}$$

$$g'_{\text{time}}(0, t) = \left(\frac{\pi}{L}\right)^{2\Delta} \left[\sinh\left(\frac{\pi\alpha t}{L}\right) \right]^{-2\Delta}. \tag{2}$$

At the matching time t_* , the unknown scaling dimension cancels:

$$g'_{\text{time}}(0, t_*) = g'_{\text{space}}(L/2, 0) \implies \boxed{\alpha = \frac{\ln(1 + \sqrt{2})L}{\pi t_*}}. \quad (3)$$

This is the prescription in the supplemental ‘‘Anisotropy parameter’’ section of Zabalo *et al.*, Phys. Rev. Lett. **128**, 050602 (2022), [arXiv:2107.03393](https://arxiv.org/abs/2107.03393).

The numerical proxy for g' is the mutual information between two reference systems that were locally entangled with the circuit at the two spacetime points. The spatial insertion uses $(\delta y, \delta \tau) = (L/2, 0)$; the temporal insertion uses $(0, \delta \tau)$. The same definition of the reference observable must be used on both legs of the match.

3 Charge-conserving Gaussian reference construction

3.1 Conventions

Let P denote the physical system and R a spectator reference. For complex fermions $\hat{\psi}_i$, use the project correlation matrix

$$G_{ij} = \text{Tr}(\hat{\rho} \hat{\psi}_i^\dagger \hat{\psi}_j). \quad (4)$$

A pure, charge-conserving Gaussian state is stored as an orthonormal occupied frame F . With the row convention used by the code, $G = F^* F^\top$, while the transposed one-body matrix materialized by the frame is $F F^\dagger$; the two have identical subsystem spectra. The circuit has two canonical physical modes $\hat{c}_{r,\mu}$, $\mu = 0, 1$, per unit cell.

3.2 One physical mode and one reference mode

For a normalized physical mode $\hat{f}$, first measure $\hat{f}^\dagger \hat{f}$ with Born probability $p = \langle \hat{f}^\dagger \hat{f} \rangle$, obtaining $m \in \{0, 1\}$. Measurement factorizes this mode from the remaining system. Append a reference mode $\hat{r}$ in the complementary occupation $1 - m$ and apply a 50:50 number-conserving beam splitter,

$$\begin{pmatrix} \hat{f}' \\ \hat{r}' \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} \hat{f} \\ \hat{r} \end{pmatrix}. \quad (5)$$

Before Eq. (5), the pair contains exactly one fermion. After it, the pair is a one-particle Bell state, up to an outcome-dependent phase. Its correlation matrix has diagonal entries $1/2$ and eigenvalues $0, 1$; the reference occupation is $1/2$ and $S(R) = \ln 2$. Thus both possible measurement outcomes yield the same amount of entanglement, no outcome is postselected, Gaussianity is preserved, and total charge on $P \cup R$ is conserved.

3.3 A full wall unit cell

The local Hilbert space of one unit cell contains two fermion modes and has dimension four. We apply the construction independently to $\mu = 0, 1$. Consequently each reference subsystem contains two modes, the freshly inserted cell/reference state is maximally entangled across this four-level cut, and

$$S(R_{\text{fresh}}) = 2 \ln 2. \quad (6)$$

Using both canonical modes avoids choosing one orbital flavor as a post-hoc definition of the local operator.

3.4 Reference mutual information

If A is any collection of reference rows, the code forms the transposed reduced correlation matrix $\tilde{G}_A = F_A F_A^\dagger$. If n_k are its eigenvalues (equivalently those of G_A), then

$$S(A) = - \sum_k [n_k \ln n_k + (1 - n_k) \ln(1 - n_k)]. \quad (7)$$

The measured correlator is

$$I(R_1 : R_2) = S(R_1) + S(R_2) - S(R_1 R_2). \quad (8)$$

Only 2×2 and 4×4 reduced correlation matrices are diagonalized. No full covariance history or Fock-space density matrix is required.

4 Circuit protocol and implementation

For every independent trajectory and every probe configuration:

1. Run the canonical CPU dynamics to $\tau_1 = 4L$.
 2. At one programmed wall and a uniformly sampled y_1 , insert R_1 using the two-mode construction above.
 3. Insert R_2 either at $(y_1 + L/2, \tau_1)$ for the spatial correlator or at $(y_1, \tau_1 + \delta\tau)$ for the temporal correlator.
 4. Continue the same Born trajectory for $2L$ additional cycles after the second insertion. Save $I(R_1 : R_2)$, $S(R_1)$, $S(R_2)$, and $S(R_1 R_2)$ at relative follow cycles $u = 0, 1, \dots, 2L$.
 5. Define the pilot plateau estimator as the average over $u = L + 1, \dots, 2L$.
- The model is the current explicit-interface production geometry:

Quantity	Pilot setting
Transverse size	$N_x = 20$
Wall circumference	$L = N_y = 6, 8$
Walls	canonical DW_loc = (4, 16)
OW range	$n_{\text{shell}} = 1$
Mass parameters	$\alpha_1 = 1, \alpha_2 = 30$
Schedule	random permutation per cycle
Feedback	perfect correction
Initialization	default pure physical frame
Samples	$S = 4$ per spatial/temporal configuration
Temporal grid	$\delta\tau = 1, \dots, 5$ for $L = 6$; $1, \dots, 6$ for $L = 8$

Every trajectory calls `classA_U1FGTN.run_markov_circuit(...)`. The pilot extends the live occupied frame by four spectator rows through the native-cycle observer. All subsequent overcomplete-Wannier measurements act only on the original $2N_x N_y$ system rows; correlations with the spectator rows nevertheless update automatically through the occupied frame. The final dimension is $2N_x N_y + 4$.

This live-state insertion is deliberately isolated in the CPU pilot. Before a production campaign, it should become an explicit reference-insertion hook in the canonical engine, with the same algebra and tests, rather than remain an observer side effect.

5 Validation

Five focused tests pass.

- Empty and occupied input modes both produce a Bell pair with pair correlation eigenvalues 0, 1, occupations $1/2, 1/2$, and reference entropy $\ln 2$.
- Two freshly and independently inserted two-mode references have $I(R_1 : R_2) = 0$, while $S(R_1) = S(R_2) = 2 \ln 2$ and $S(R_1 R_2) = 4 \ln 2$.
- Linear matching recovers a known synthetic t_* and Eq. (3).
- A nonphysical upward noise excursion cannot create a late crossing.
- The canonical engine continues after all four spectator modes are added; the augmented frame remains orthonormal and all reference observables are finite and nonnegative.

The separate end-to-end smoke run and the production-size pilot both completed without missing cycles. The augmented-frame Gram residual remained below the hard 10^{-8} check (the direct integration smoke gave approximately 10^{-15}).

6 Estimator and statistical rules

The two walls are balanced across sample indices, but they are not promoted to independent samples. A complete Born trajectory is the statistical unit. Spatial and temporal configurations use matched sample/seed identities, and the nonparametric bootstrap resamples these identities jointly.

The mean temporal curve must begin above the spatial target and cross it downward as $\delta\tau$ increases. If the shortest sampled temporal point is already below the target, a later upward fluctuation followed by a downward fluctuation is rejected. Within a valid adjacent bracket, t_* is found by linear interpolation and inserted in Eq. (3). This rule is more restrictive than merely finding any sign change, and prevents rare-trajectory noise from manufacturing an anisotropy.

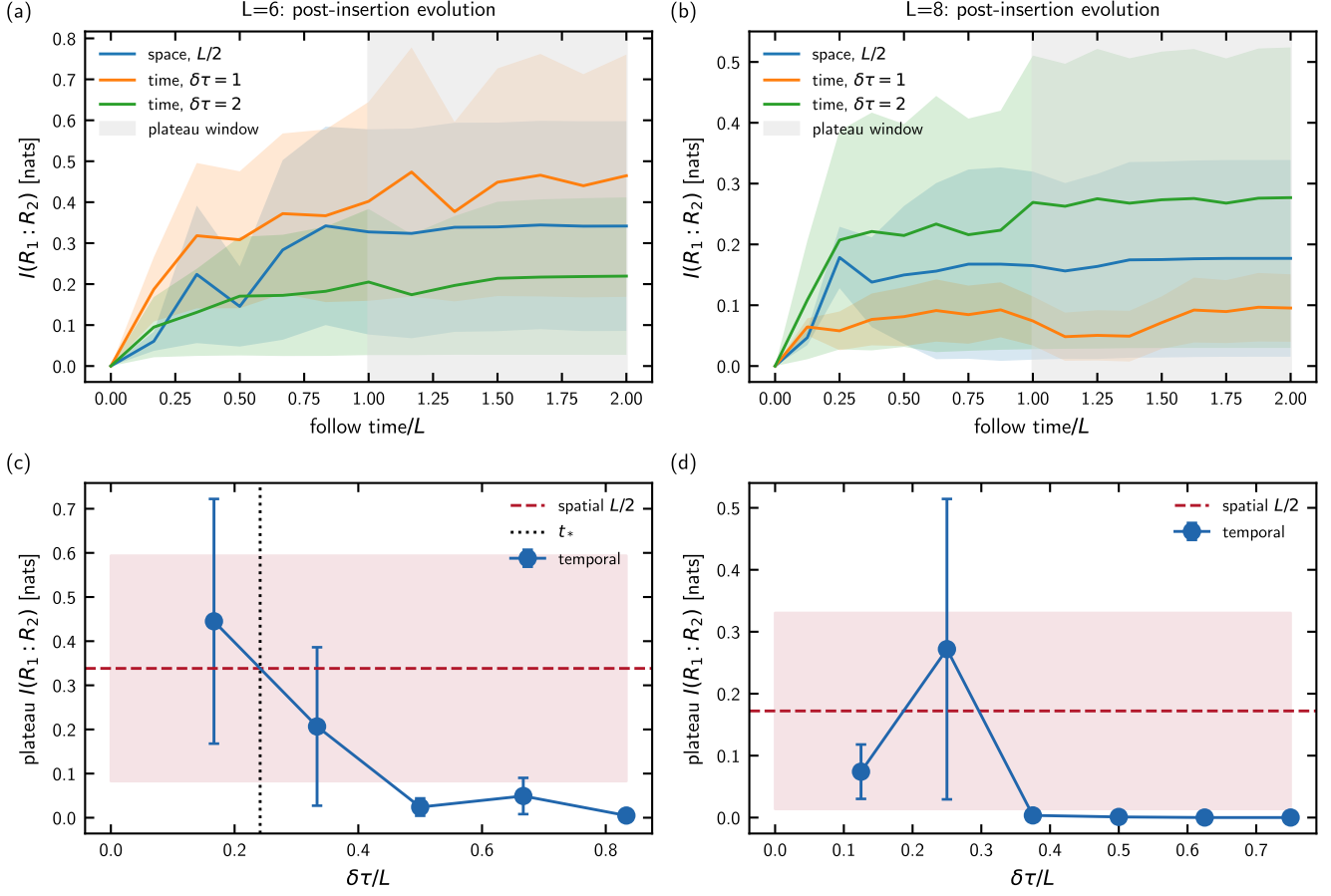


Figure 1: Reference-ancilla CPU pilot. Top: post-insertion mutual-information evolution; shading is the trajectory standard error and gray marks the final L -cycle plateau window. Bottom: plateau temporal correlators and the spatial $L/2$ target. For $L = 6$ the mean has an ordered candidate crossing between $\delta\tau = 1, 2$. For $L = 8$ the large $\delta\tau = 2$ point is supplied mainly by one trajectory; because the $\delta\tau = 1$ mean is already below the spatial target, the strict analysis rejects the apparent later crossing.

7 Pilot results

The temporal plateau means are

$$\begin{aligned}
 L = 6 : & \quad (0.4450, 0.2067, 0.0241, 0.0492, 0.00492), \\
 L = 8 : & \quad (0.0741, 0.2719, 0.00350, 0.00104, 7.54 \times 10^{-5}, 8.03 \times 10^{-5}).
 \end{aligned}$$

The $L = 8$ ordering violation is visible directly: the second point is larger than the first. At the trajectory level, one of four samples contributes 0.9973 to the $\delta\tau = 2$ plateau, while the other three contribute 2.9×10^{-6} , 8.8×10^{-3} , and 8.15×10^{-2} . Likewise, one trajectory contributes 0.6458 of the $L = 8$ spatial signal while two are near 10^{-6} . This is the broad, rare-event-dominated distribution expected for reference mutual information in monitored dynamics; a four-sample mean is not stable.

The plateau diagnostic compares the preceding and final L -cycle windows. For the configurations that control the candidate bracket, the paired changes are

Table 1: Strict matching results. Bootstrap resolution is the fraction of whole-trajectory resamples that contain an ordered downward bracket.

L	$\bar{I}_{\text{space}}$	t_*	α	bootstrap resolution	estimated S for 20% spatial SEM
6	0.3383	1.448	1.163 (candidate)	0.756	57
8	0.1721	unresolved	unresolved	0.266	85

Configuration	previous L -window	final L -window; paired change
$L = 6$, spatial	0.2305	0.3383; $+0.1078 \pm 0.0977$
$L = 6$, $\delta\tau = 1$	0.3258	0.4450; $+0.1192 \pm 0.1045$
$L = 6$, $\delta\tau = 2$	0.1595	0.2067; $+0.0472 \pm 0.0448$
$L = 8$, spatial	0.1461	0.1721; $+0.0260 \pm 0.0602$
$L = 8$, $\delta\tau = 1$	0.0778	0.0741; -0.0037 ± 0.0188
$L = 8$, $\delta\tau = 2$	0.2117	0.2719; $+0.0602 \pm 0.0557$

Errors here are standard errors of the four paired trajectory changes. The $L = 6$ controlling curves still rise by roughly one standard error, reinforcing the decision to call its crossing a candidate rather than a calibration.

8 Why the earlier record-field pilot failed

Before implementing references, a surrogate pilot compared connected local record surprisal and mismatch activity in space and time. Those observables had a large lag-zero contact term and noise-scale correlations at every nonzero lag and at $L/2$. Interpolating from lag zero to lag one produced fictitious sub-cycle crossings. Once the contact term was excluded, all cases were unresolved. This was an estimator mismatch, not evidence that the wall lacks an anisotropic critical sector. The present reference pilot confirms the distinction: it produces a nonzero, evolving, long-lived mutual-information signal.

9 Recommended next CPU campaign

9.1 How the necessary cycle count is determined

The three contributions to trajectory length answer different questions and must not be conflated:

$$T_{\text{total}} = T_{\text{eq}} + \delta\tau_{\text{max}} + T_{\text{follow}}. \quad (9)$$

T_{eq} removes dependence on the initial Gaussian state; $\delta\tau_{\text{max}}$ is only the temporal distance needed to bracket t_* ; and T_{follow} allows the two-reference mutual information to reach its late-time value after the second insertion.

For the support-terminated geometry, determine these terms with matched Born seeds as follows.

1. Run an equilibration grid $T_{\text{eq}}/L = 2, 3, 4, 6$ while holding a long $T_{\text{follow}} = 3L$. For the spatial point and the two temporal points bracketing the crossing, compare adjacent equilibration choices using the paired trajectory differences. Accept the smallest T_{eq} for which the 95% paired-bootstrap interval contains zero, the point shift is smaller than half the final standard error, and the inferred t_*/L is stable under the next larger choice.
2. In a run followed through at least $3L$, form consecutive nonoverlapping L -cycle window means after the second insertion. Accept the smallest T_{follow} for which the last two windows pass the same paired zero-shift and half-standard-error criteria for the spatial point and both bracketing temporal points. If no pair passes, extend the follow to $4L$ rather than broadening the averaging window until drift is hidden.
3. Increase $\delta\tau$ only until the mean temporal correlator has an ordered downward bracket around the spatial target. Once a preliminary anisotropy is available, the expected location is

$$\frac{t_*}{L} = \frac{\ln(1 + \sqrt{2})}{\pi\alpha} \simeq \frac{0.28055}{\alpha}. \quad (10)$$

Add every integer cycle inside that bracket; points far beyond it do not improve the interpolation.

The audit must be performed on the same support-terminated protocol and cycle definition as production. Copying Zabalo's $4L$ equilibration from a different circuit, or copying the untruncated pilot, does not establish convergence.

The sample coefficient of variation is 1.51 ($L = 6$ spatial) and 1.84 ($L = 8$ spatial). The elementary estimate

$$S_{20\%} \simeq \left(\frac{CV}{0.20} \right)^2 \quad (11)$$

gives 57 and 85 trajectories merely to reach a 20% relative standard error on the spatial mean. Temporal configurations in the scanned range require approximately 31–100 trajectories by the same diagnostic. Therefore:

1. use at least $S = 100$ trajectories per configuration for the next calibration pilot; $S = 200$ is preferable before quoting a production uncertainty;
2. retain $L = 6, 8$ and add $L = 10$ only after the first two sizes show a stable ordered bracket;
3. keep $\tau_1 = 4L$ and a $2L$ post-insertion follow initially, but require agreement between the preceding and final L windows; extend to $3L$ if that test fails;
4. use a coarse temporal grid through $0.75L$, then add every integer cycle only inside the resolved bracket;
5. report the arithmetic Born mean and the full trajectory distribution; do not replace the mean by a typical or median mutual information.

For the present protocol, the largest trajectory length is

$$T_{\text{aniso}} = 4L + 0.75L + 2L = 6.75L, \quad (12)$$

so allocating $7L$ cycles per reference-anisotropy trajectory is appropriate. This is the initial cycle budget for the *calibration experiment*, not an assumed necessary duration and not yet a measured conversion factor for every other production observable. It must be checked anew with `dw_truncation=meas_slab_only=True`. Once α is stable, the natural CFT time unit is L/α and production aspect ratios can be stated as $\alpha T/L$.

The completed 52-trajectory-configuration pilot used 455.9 CPU seconds. At the observed serial rate, $S = 100$ over the same $L = 6, 8$ grid is about 3.2 CPU-hours and is straightforward to distribute across independent worker processes.

10 Artifacts and reproducibility

The implementation and outputs are contained in `00_WORKSPACE/CURRENT/reference_ancilla_anisotropy_cpu_pilot`. The key files are:

- `reference_probe.py`: Gaussian insertion, entropies, mutual information, and strict matching;
- `run_cpu_pilot.py`: canonical dynamics runner, raw retention, bootstrap, and figures;
- `reanalyze_output.py`: lossless strict reanalysis of a completed run;
- `tests/`: analytic and canonical integration tests;
- `outputs/pilot_20260826_183458/`: manifest, complete raw NPZ arrays, bootstrap arrays, JSON results, summary, and figure.

Each raw configuration file retains, with logical leading axes (sample, follow cycle), the mutual information and all three component entropies. It also stores both insertion sites, both orbital outcomes and Born probabilities, engine and probe seeds, absolute cycle coordinates, and the canonical dynamics entry point. The manifest records source hashes and the dirty Git state. The superseded permissive interpolation is preserved as a diagnostic and is not silently overwritten as an accepted result.

Reproduce the validation and run with

```
pytest -q 00_WORKSPACE/CURRENT/reference_ancilla_anisotropy_cpu_pilot/tests
python 00_WORKSPACE/CURRENT/reference_ancilla_anisotropy_cpu_pilot/run_cpu_pilot.py
```